# Applied Statistics

A.A. 2025/2026

## LINEAR MODELS – Part 1

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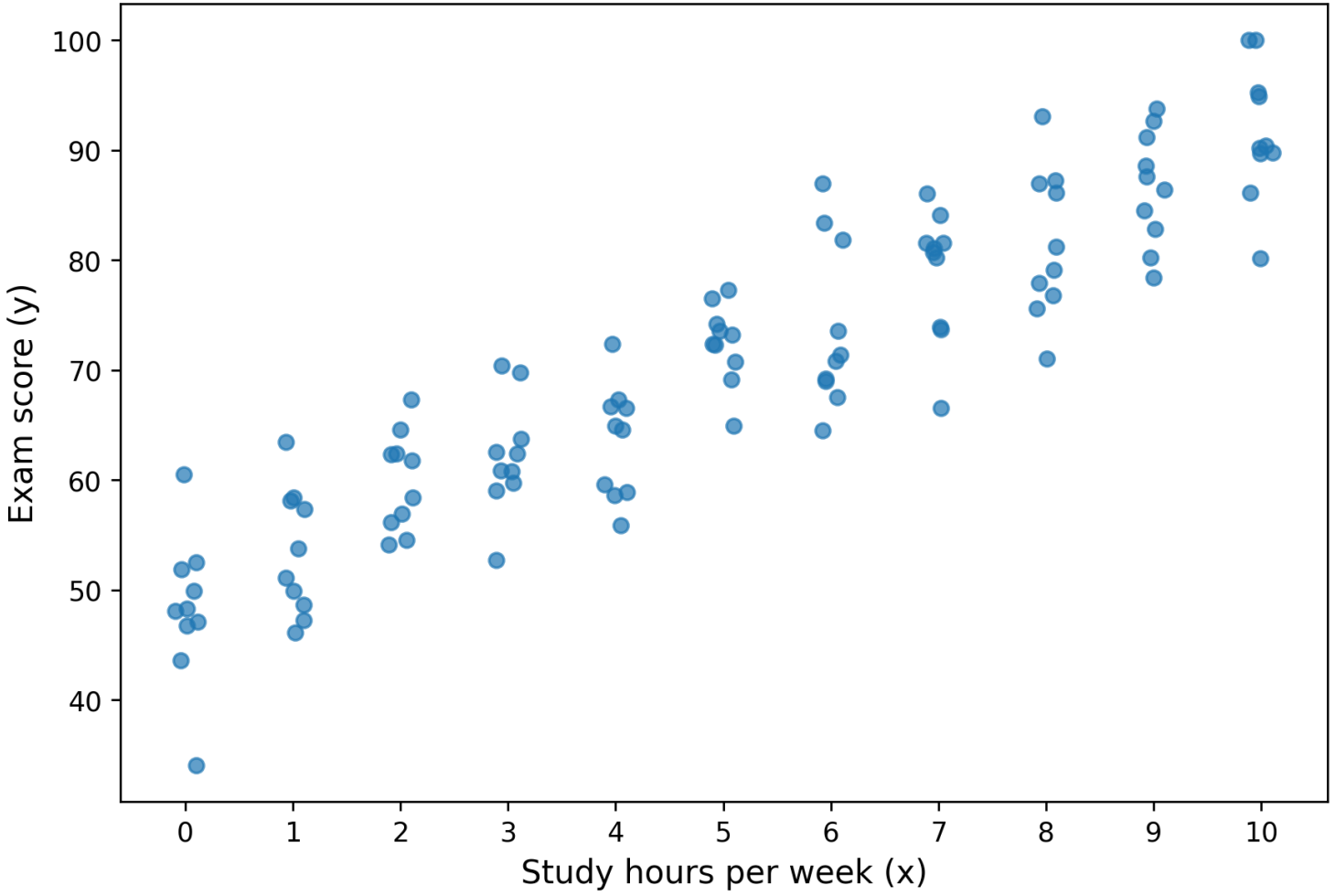
# The simple linear regression model

# First example: exam score prediction



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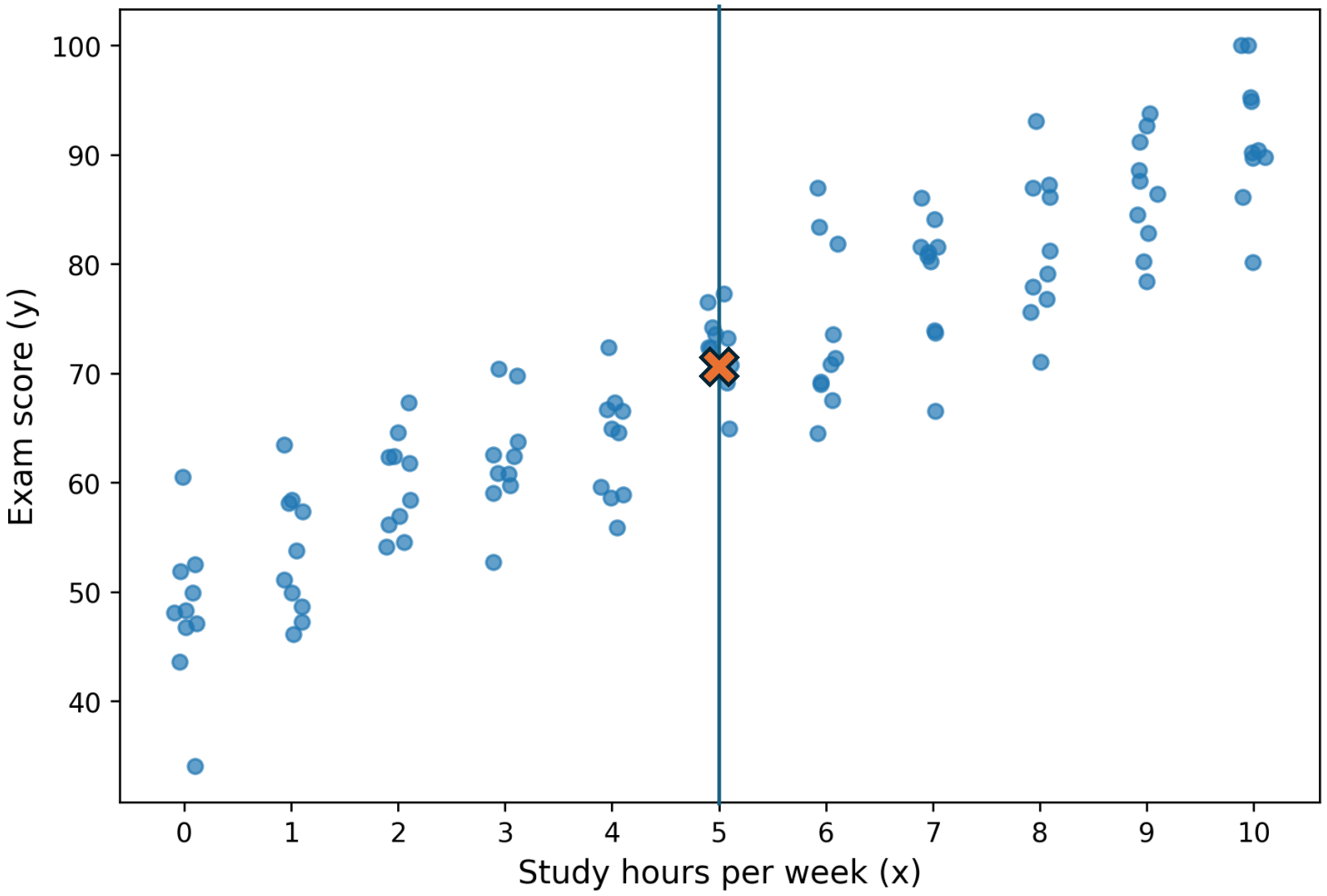

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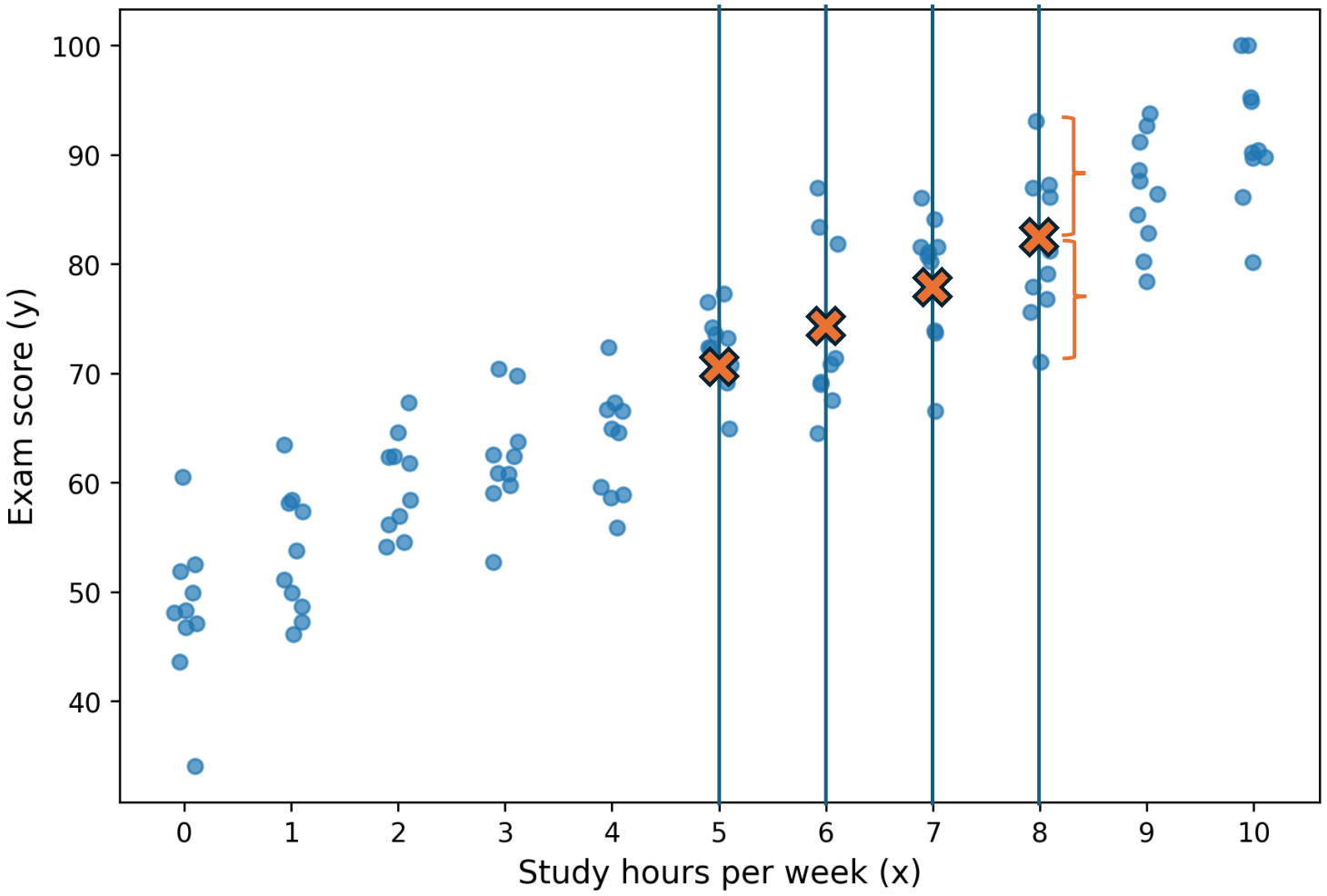

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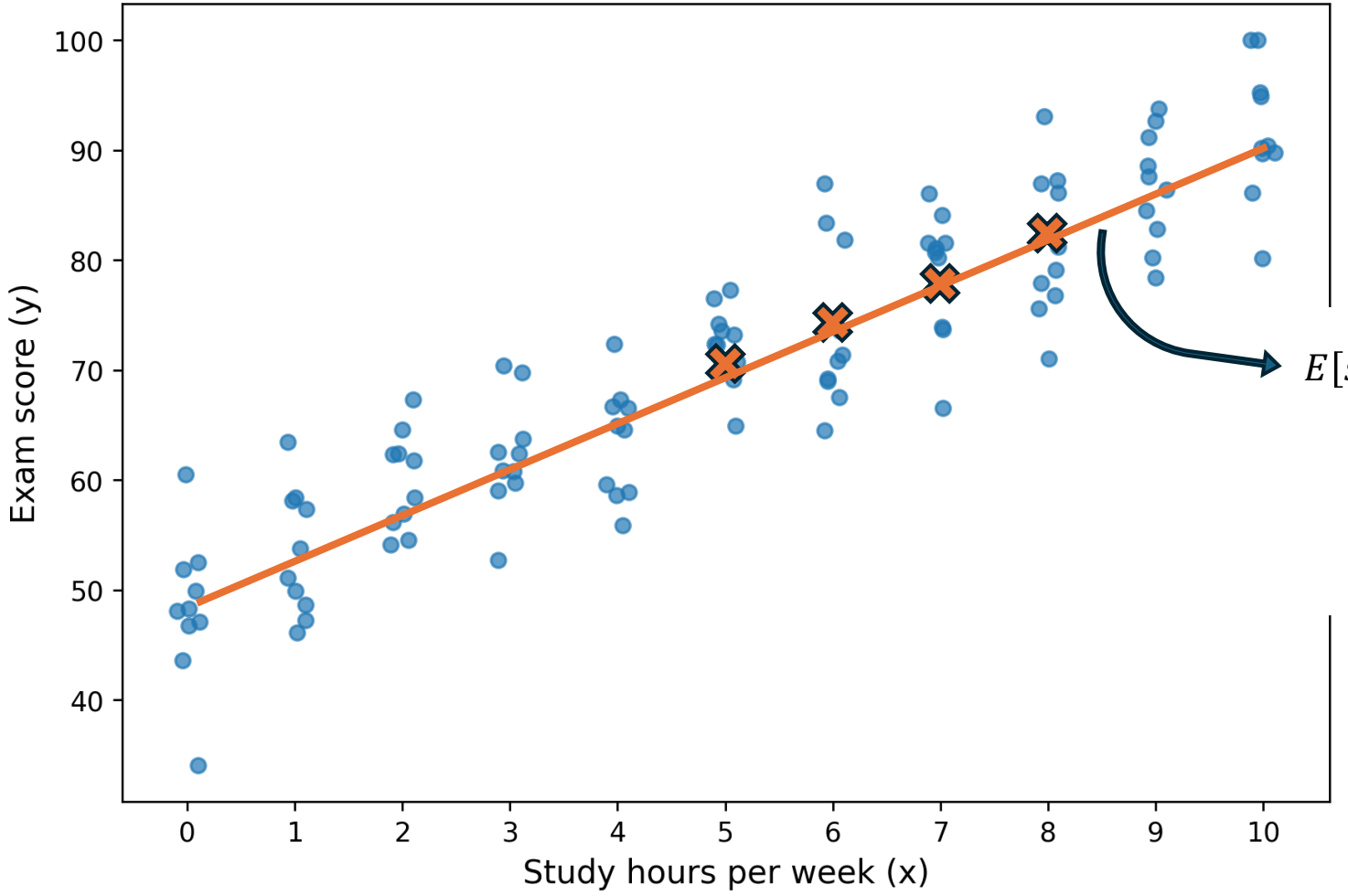


$$E[score|study\ hours] = f(study\ hours)$$

$$E[Y|X] = f(X)$$

$$E[Y|X] = \beta_0 + \beta_1 X + \varepsilon$$

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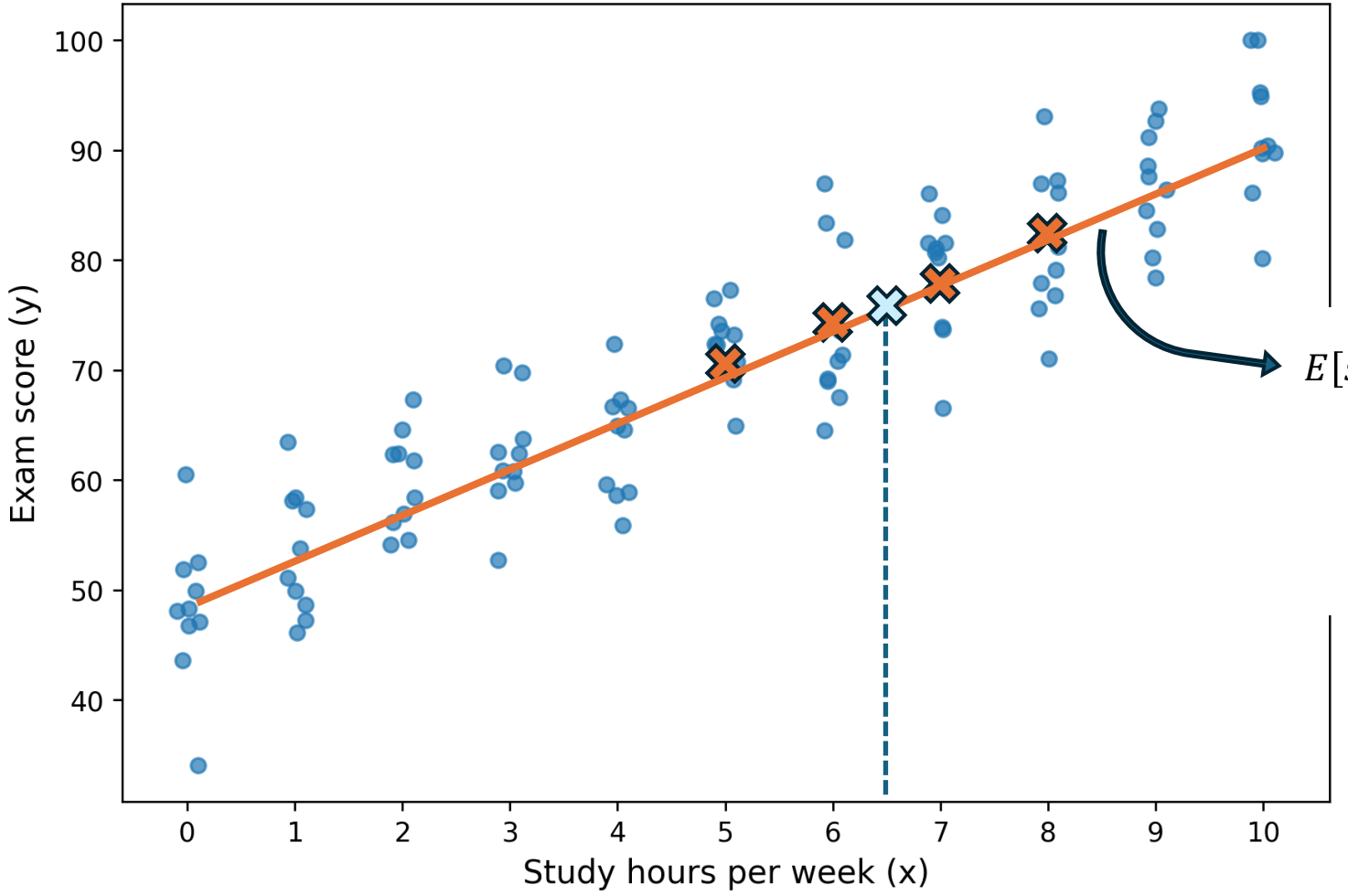


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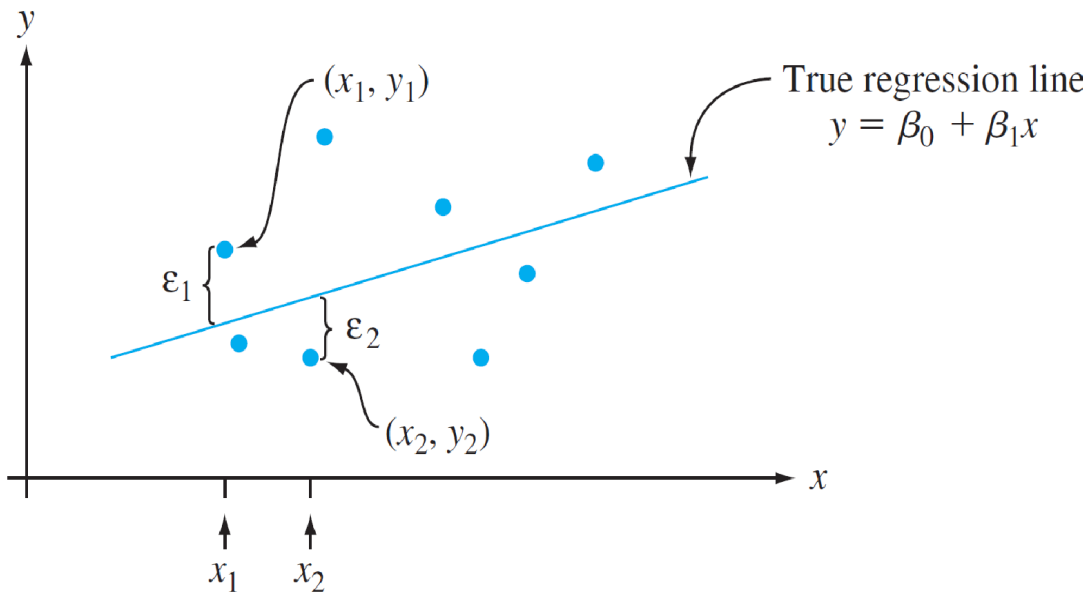
# The simple linear regression model

*X*: the **independent, predictor,** or **explanatory variable** (usually known).

*Y:* The **dependent** or **response variable.** For **fixed *x***, ***Y*** will be <u>random variable</u>.

*ε:* The **random deviation** or **random error term**. For **fixed *x*, *ε*** will be <u>random variable</u>.

If we think of an entire population of $(x, y)$ pairs, then $\mu_{Y|x*}$ is the *mean of all y values for which x=x**, and $\sigma^2_{Y|x*}$ is a measure of how much these values of y spread out about the mean value.



# The simple linear regression model

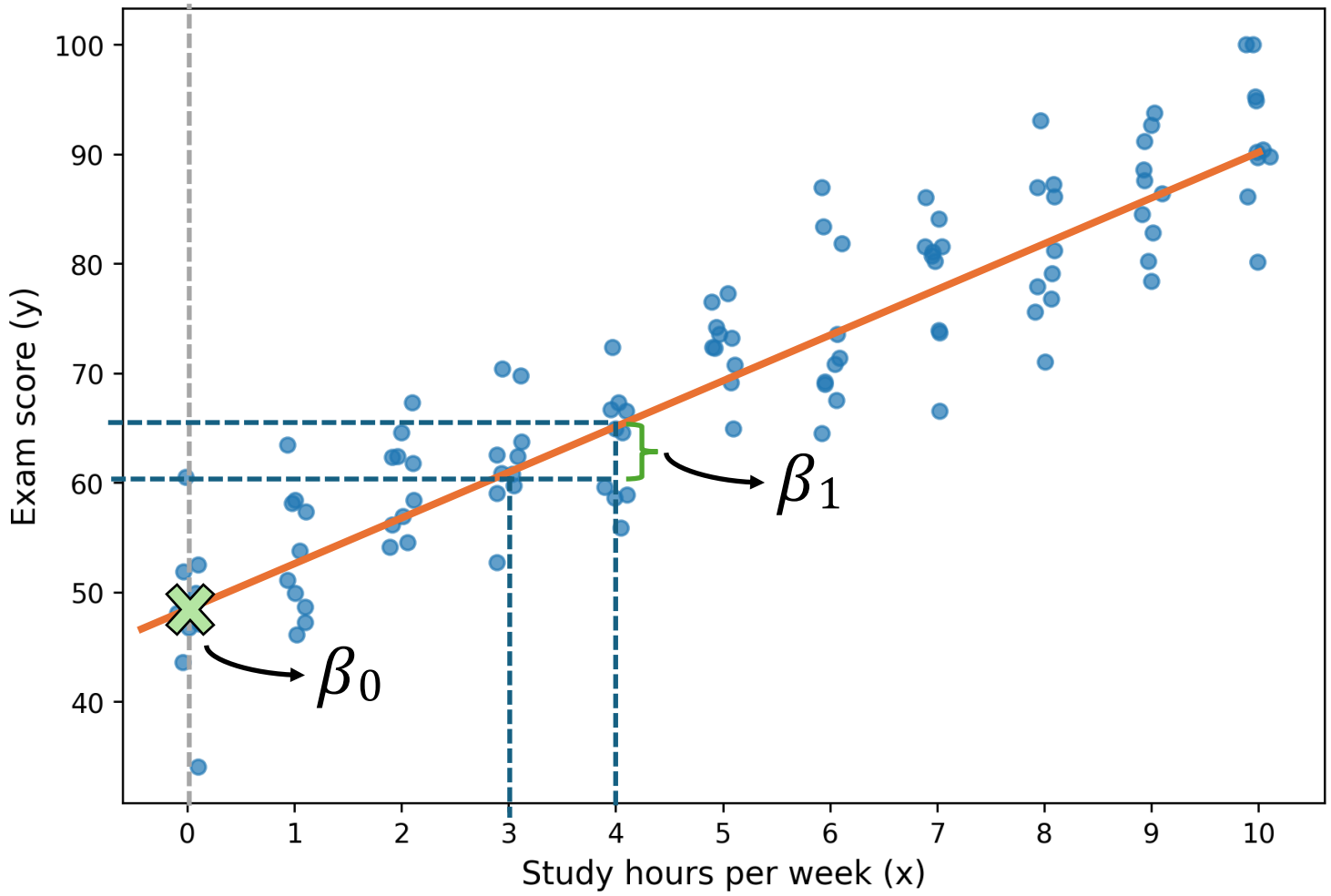
There are parameters, $\beta_0$, $\beta_1$, and $\sigma^2$ such that *for any fixed value of the independent variable x, the dependent variable is a random variable* related to *x* through the **model equation**

$$Y = \beta_0 + \beta_1 x + \varepsilon$$

The quantity $\varepsilon$ in the model equation is the "error": a <u>random variable</u>, assumed to be symmetrically distributed with $E[\varepsilon] = 0$ and $V[\varepsilon] = \sigma_\varepsilon^2 = \sigma^2$

(no assumptions on the error yet)

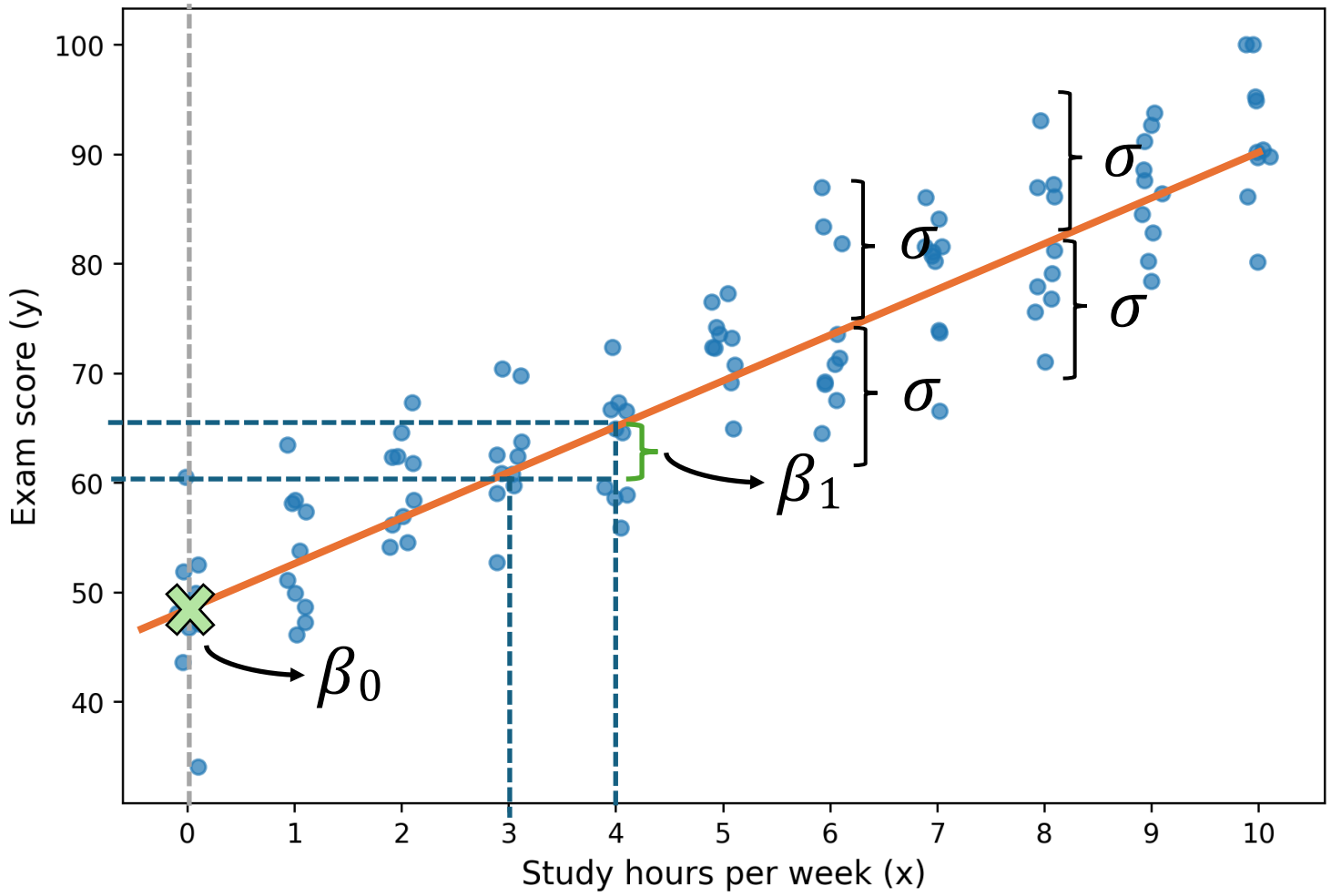
# The simple linear regression model: parameters




$\beta_0$ – the intercept of the true regression line: the average value of Y when x is zero.

$\beta_1$ – the slope of the true regression line: the expected (average) change in Y associated with a unit increase in the value of x

# The simple linear regression model: parameters

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$\sigma^2 = V[\varepsilon]$

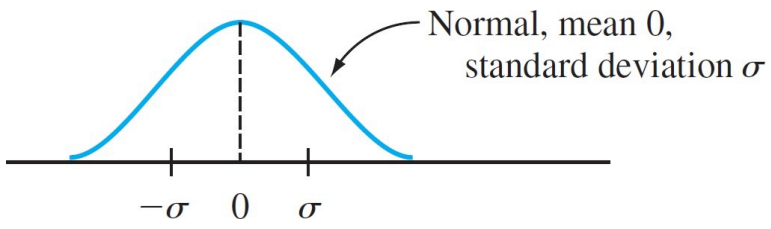
# The simple linear regression model: parameters

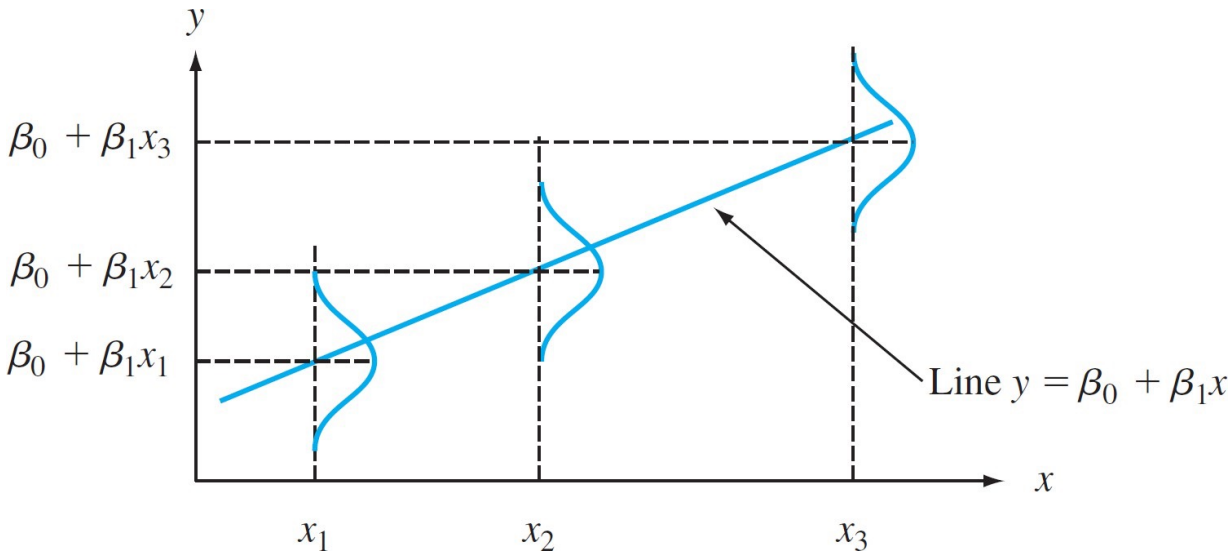
- We assume the variance (amount of variability) of the distribution of *Y* values to be the same at each different value of fixed *x*. (*i.e.* homogeneity of variance assumption: **homoscedasticity**).
- We also assume that $\boldsymbol{\varepsilon \sim \mathcal{N}(0, \sigma^2)}$ **- gaussianity.**

The variance parameter $\sigma^2$ determines the extent to which each normal curve spreads out about the regression line.

→ Small value for $\sigma^2$ means that an observed point $(x, y)$ will almost always fall quite close to the regression line.

→ When $\sigma^2$ is large, observations may deviate considerably from their expected values.



# Finding model parameters

The values of $\beta_0$, $\beta_1$, and $\sigma^2$ are usually not known to an investigator.

Instead, sample data consists of n observed pairs $(x_1, y_1), \dots, (x_n, y_n)$ from which the model parameters and the regression line itself can be **estimated**.

We assume that

$$y_i = \beta_0 + \beta_1 x_1 + \varepsilon_i$$

And

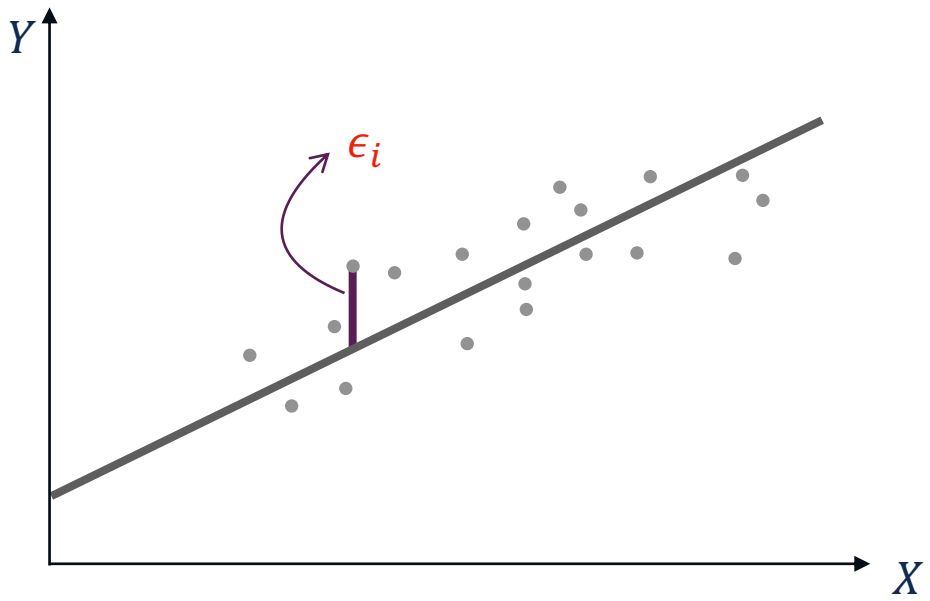
$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

- For $i = (1, \dots, n)$ and the n deviations $\varepsilon_1, \dots, \varepsilon_n$ are **independent random variables**.
- **Y is random**, modeled by the data $y_i | x_i \sim \mathcal{N}(\beta_0 + \beta_1 x_1, \sigma^2)$
- $x$ is known and deterministic
- $\beta$ is unknown and deterministic

# Parameter estimation: least squared method

We look for $\beta$ that minimize the squared errors between data and the model:

$$\widehat{\beta_{LS}} = \underset{\beta}{\operatorname{argmin}} \sum_i \varepsilon_i^2 = \sum_i (y_i - [\beta_0 + \beta_1 x_1]^2)$$



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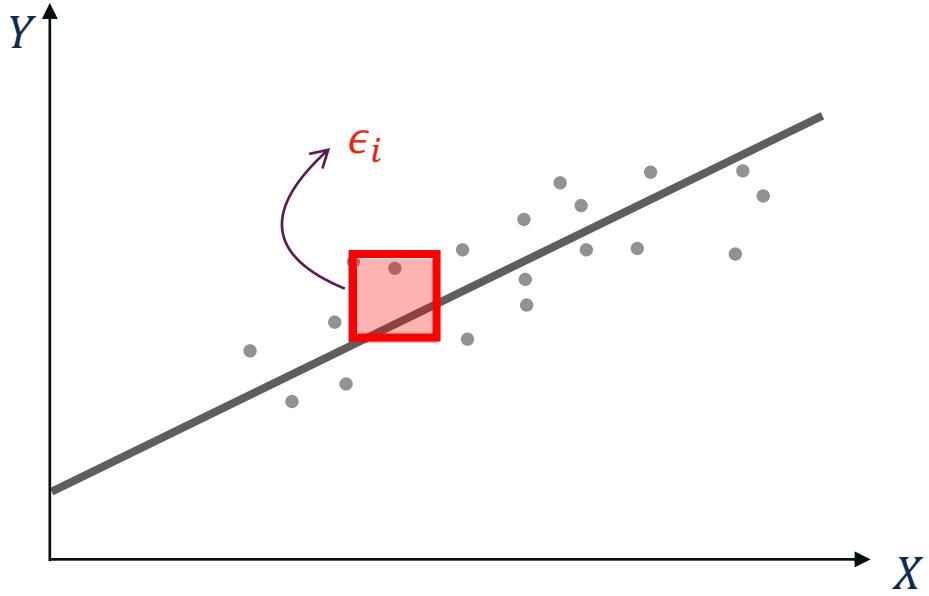

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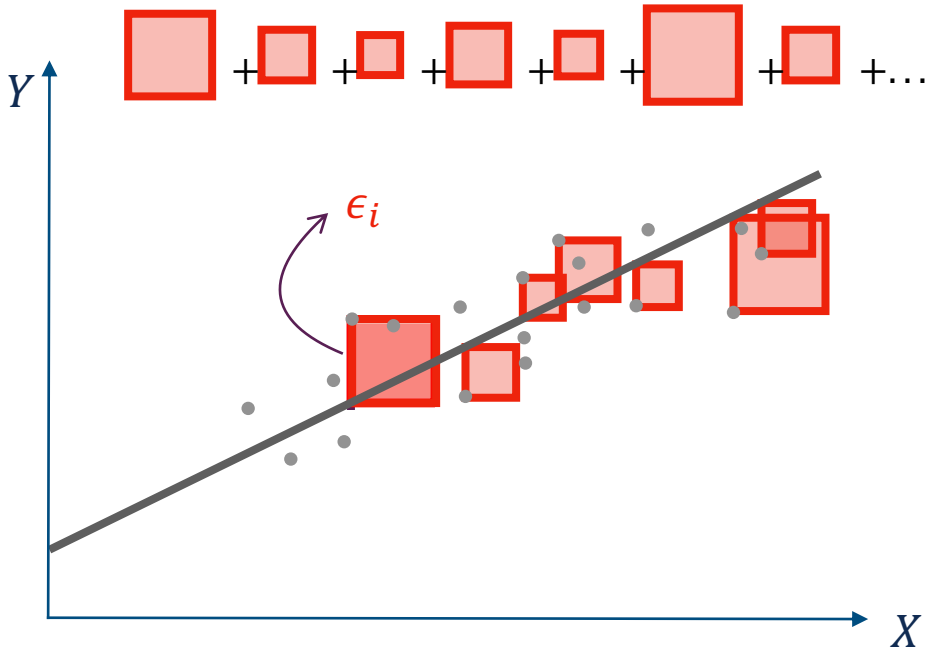

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**Solution.** not important

$$\begin{cases} \frac{\partial}{\partial \beta_0} \sum_i (y_i - [\beta_0 + \beta_1 x_1]^2) = -2 \sum_i (y_i - \beta_0 - \beta_1 x) = 0 \\ \frac{\partial}{\partial \beta_1} \sum_i (y_i - [\beta_0 + \beta_1 x_1]^2) = -2 \sum_i (y_i - \beta_0 - \beta_1 x) = 0 \end{cases} \Longrightarrow$$

$$\begin{cases} \sum_i y_i = n\beta_0 + \beta_1 \sum_i x_i \\ \sum_i y_i x_i + \beta_0 \sum_i x_i - \beta_1 \sum_i x_i = 0 \end{cases} \Longrightarrow \begin{cases} \bar{y} = \beta_0 + \beta_1 \bar{x} \\ \sum_i x_i^2 \beta_1 = \sum_i y_i x_i - (\bar{y} - \beta_1 \bar{x}) \sum_i x_i \end{cases}$$

$$\begin{cases} \bar{y} = \beta_0 + \beta_1 \bar{x} \\ \beta_1 \left( \sum_i x_i^2 - \frac{(\sum_i x_i)^2}{n} \right) = \sum_i x_i y_i - \frac{\sum_i x_i y_i}{n} \end{cases} \Longrightarrow \begin{cases} \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \\ \hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \end{cases}$$

where $\bar{x} = \frac{1}{n} \sum_{I=1}^{n} x_i$ and $\bar{y} = \frac{1}{n} \sum_{I=1}^{n} y_i$

sample means

# Parameter estimation: inference about model parameters

The least squares method finds parameters that satisfy

$$\begin{cases} \hat{\beta}_1 = \dfrac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \\ \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \end{cases}$$

*How good is this estimate?*

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*How good is this estimate?*

The standard error (variance) of an estimator reflects how it varies under repeated sampling. We have:

$$\begin{cases} SE(\hat{\beta}_0)^2 = Var(\hat{\beta}_0) = \sigma^2 \left[ \dfrac{1}{n} + \dfrac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \right] \\ SE(\hat{\beta}_1)^2 = Var(\hat{\beta}_1) = \dfrac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \end{cases} \Longrightarrow \begin{cases} \hat{\beta}_0 \sim \mathcal{N}(\beta_0, SE(\hat{\beta}_0)^2) \\ \hat{\beta}_1 \sim \mathcal{N}(\beta_1, SE(\hat{\beta}_1)^2) \end{cases}$$

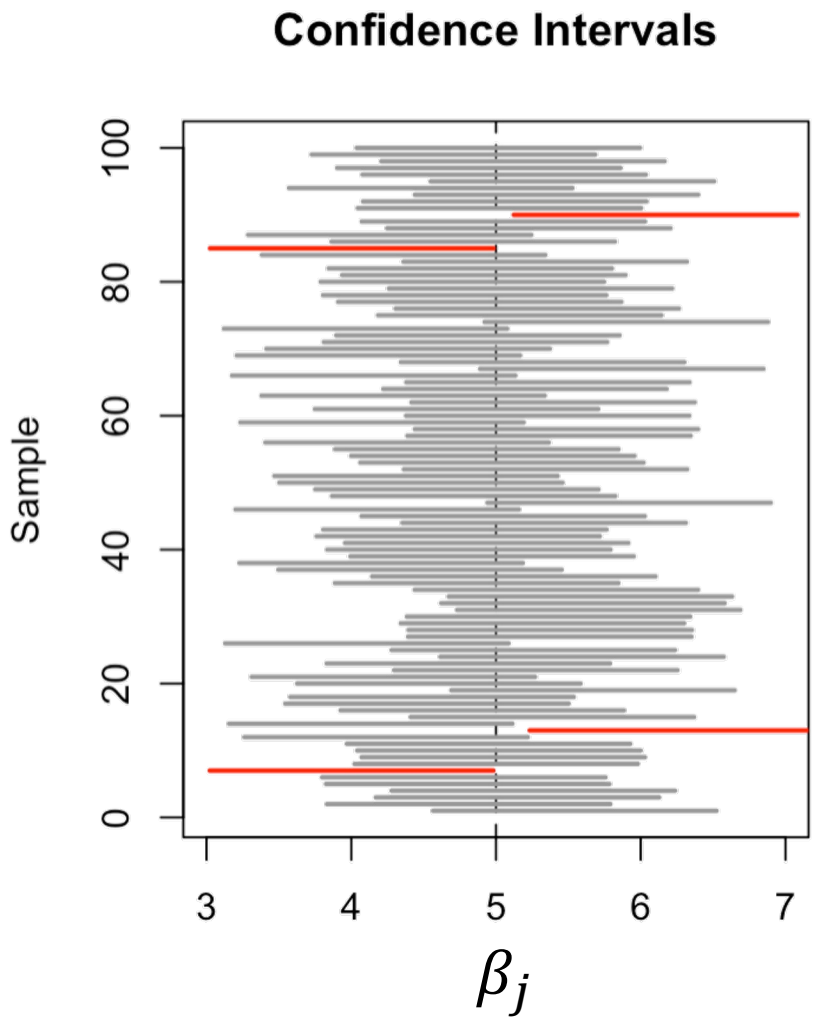
# Parameter estimation: inference about model parameters

These standard errors can be used to compute confidence intervals. A 95% confidence interval is defined as a range of values such that, with 95% probability, the range will contain the true unknown value of the parameter. It has the form:

$$\hat{\beta}_j \pm t_{n-2,\, 1-\alpha/2}\, \mathrm{SE}(\hat{\beta}_j), \qquad j \in \{0, 1\}$$

The interval has a probability of 95% to contain the true value of $\beta_j$.

- **A $95\%$ CI is a procedure that, over repeated samples, contains the true $\beta_j$ about $95\%$ of the time.**
- It is **not** "there is 95% probability that $\beta_j$ lies in this specific interval" (frequentist nuance).



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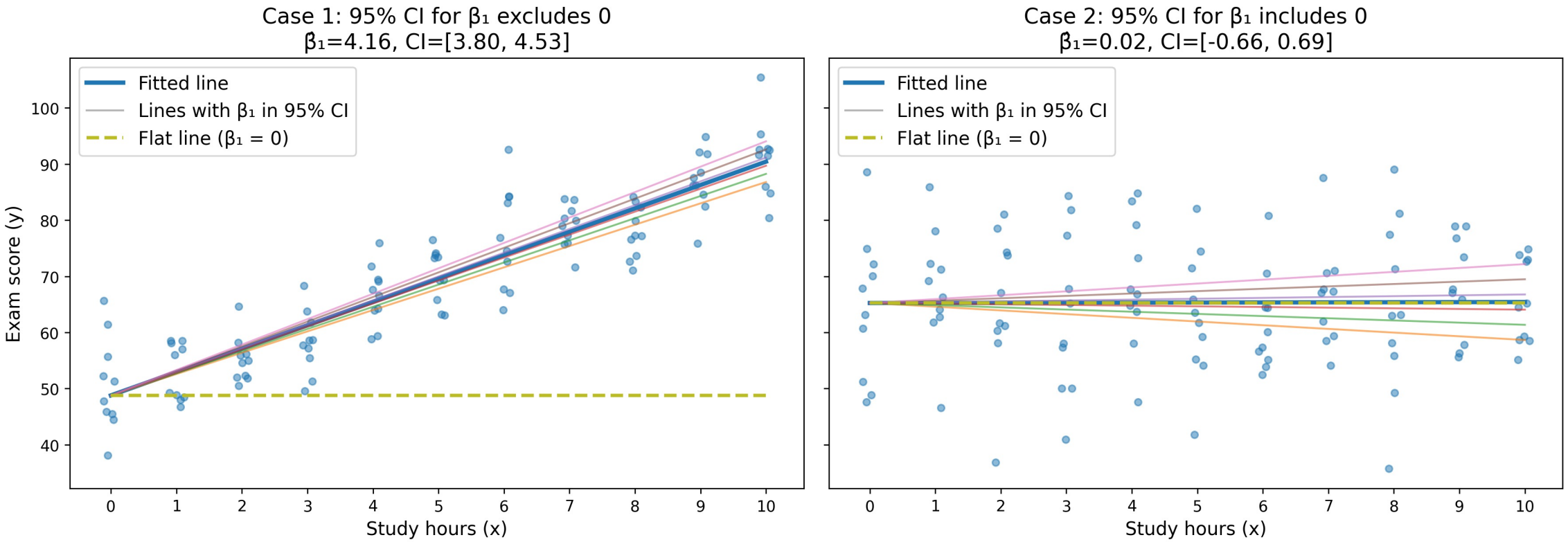
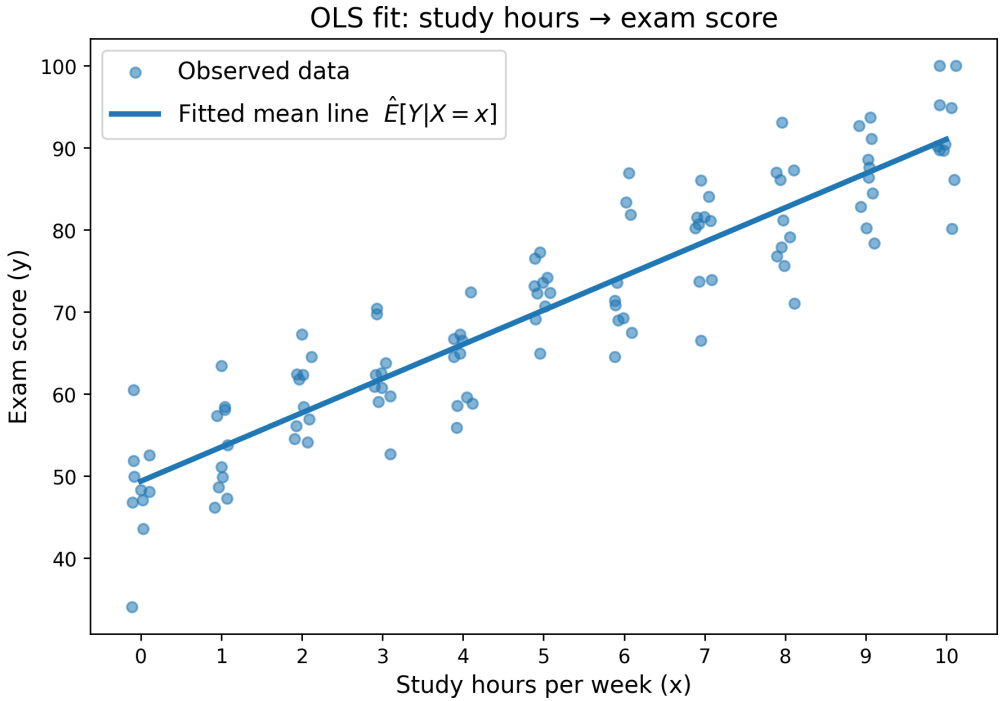


# Assessing the model

| | Estimate | Std. Error | t value | p-value | CI 2.5% | CI 97.5% |
|---|---|---|---|---|---|---|
| **Intercept ($\beta_0$)** | 49.377 | 1.010 | 48.889 | 1.690e-75 | 47.375 | 51.379 |
| **Slope ($\beta_1$)** | 4.167 | 0.1707 | 24.409 | 9.159e-46 | 3.829 | 4.506 |




| Model metrics | Value |
|---|---|
| n (observations) | 110.0 |
| Degrees of freedom resid | 108.0 |
| RSS | 3462.56 |
| $\hat{\sigma}$ (residual SD) | 5.66 |
| $R^2$ | 0.846 |
| Adjusted $R^2$ | 0.845 |
| F-statistic (overall) | 595.8135 |
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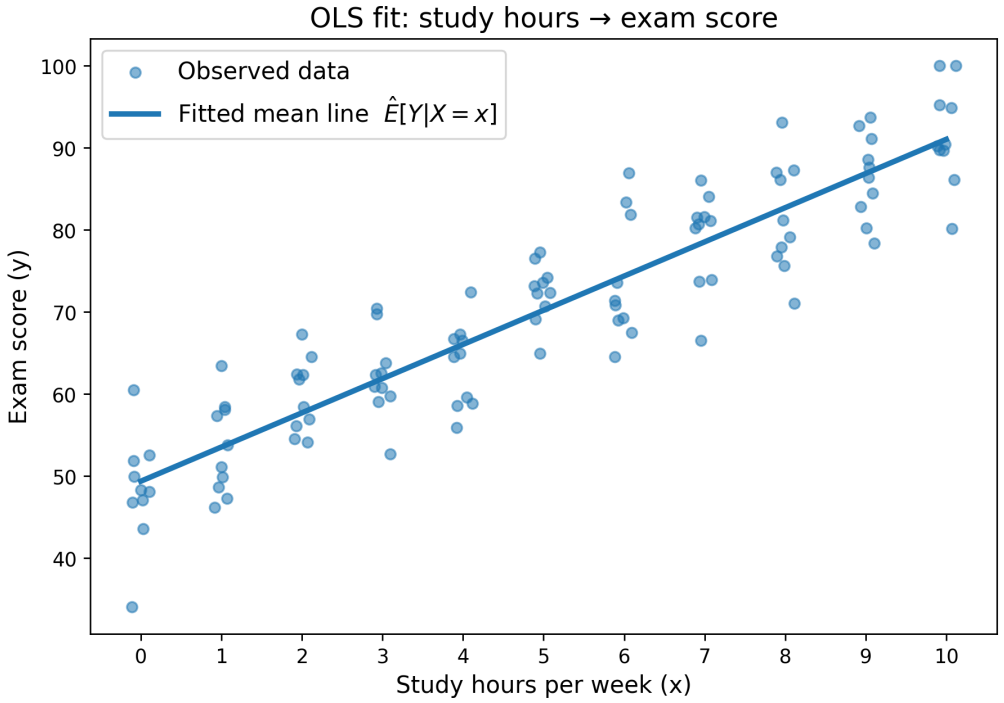


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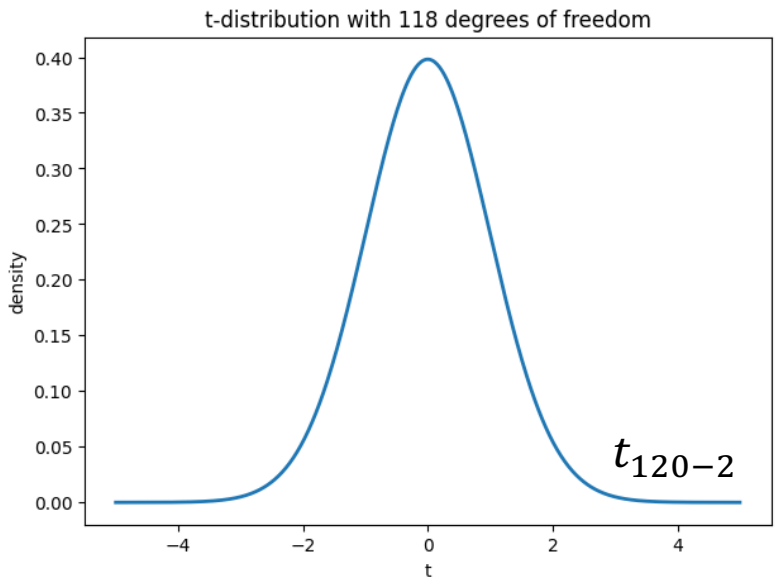
# Assessing the model

## P-value of the t-test for model's coefficients

For each coefficient $\beta_j$ (intercept and slope), the standard test is:

$$H_0: \beta_j = 0 \quad \textit{vs} \quad H_1: \beta_j \neq 0$$




**Test statistic**

$$t_j = \frac{\hat{\beta}_j}{\mathrm{SE}(\hat{\beta}_j)}$$

Under $H_0$, and assuming the classical model conditions (independent errors, constant variance, normality for exact small-sample inference),

$$t_j \sim t_{n-2}$$

**Interpretation**

- Does the regression model reduce the error *enough* compared to the baseline, beyond what could happen just by random noise?
- A **small p-value** means: *if the true coefficient were 0, getting a slope estimate this large (much higher than 0 in absolute value) would be very unlikely* → evidence that the coefficient is not zero.
- A **large p-value** means: the data are compatible with $\beta_j = 0$ (you don't have evidence of an effect).
- $p - value < 0.05$ → $0 \notin \{95\%\ \mathrm{CI}\}$

# Assessing the model: variability explained

Given observations $(x_i, y_i)$ and fitted values $\hat{y}_i$, in simple linear regression, $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, and $\hat{e}$ are the model **residuals** (different from $\varepsilon$).

$$\forall (x_i, y_i):$$
$$\hat{y}_i \to \hat{\beta}_0 + \hat{\beta}_1 x_i + e_i$$
$$y_i \to \beta_0 + \beta_1 x_i + \epsilon_i$$
$$\implies e_i \neq \epsilon_i$$

so, the Residual Sum of Squares:

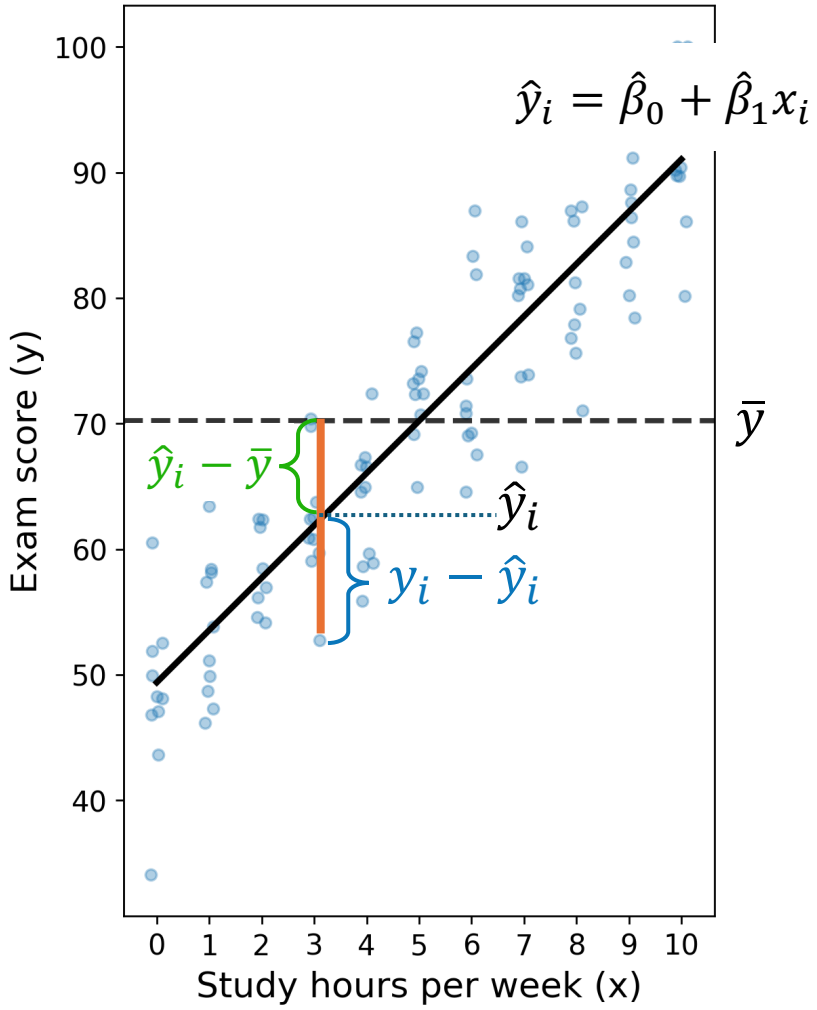
$$\mathrm{RSS} = \sum_{i=1}^{n} \left(y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right)\right)^2$$

Let $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$. Then

$$\mathrm{TSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

**Interpretation:** total variability of $y$ around its mean (a baseline model that predicts $\bar{y}$ for everyone).

# Assessing the model: variability explained




$$TSS = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

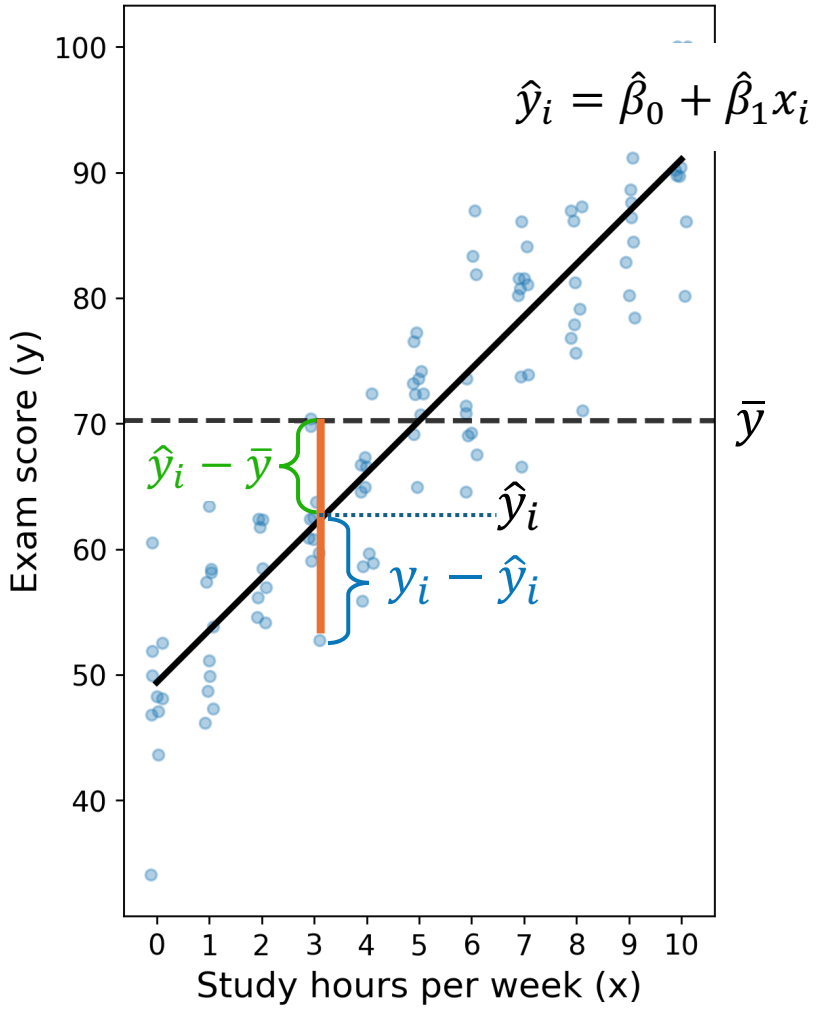
**TSS** = **ESS** + **RSS**

**TSS**: Total Sum of Squares

**ESS**: Explained Sum of Squares

**RSS**: Residual Sum of Squares

# Assessing the model: variability explained




$$TSS = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

**TSS** = **ESS** + **RSS**

Total Sum of Squares = Explained Sum of Squares + Residual Sum of Squares

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS}$$

**variance explained:** proportion of variability in $y$ explained by the regression, compared to predicting $\bar{y}$.

→ $\boldsymbol{R^2 \in [0, 1]}$

$$R^2_{adj} = 1 - \frac{RSS/(n-p)}{TSS/(n-1)}$$

With $p$ parameters. Like $R^2$, but **penalizes model complexity** (useful once you move to multiple regression).

# Assessing the model: residual variability/prediction error

### Residual standard error (typical prediction error on training data)

$$\hat{\sigma}^2 = \frac{\text{RSS}}{n-p}, \qquad \hat{\sigma} = \sqrt{\hat{\sigma}^2} \qquad \hat{\sigma}^2 = \frac{\text{RSS}}{n-2} \text{ for simple linear regression}$$

**T**ypical size of residuals in the units of $y$ (e.g., "about $\hat{\sigma}$ exam-score points").

### Root mean squared error (RMSE)

Average prediction error magnitude, with squared errors giving more weight to large mistakes. Similar to RSE, but not corrected for model degrees of freedom-

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

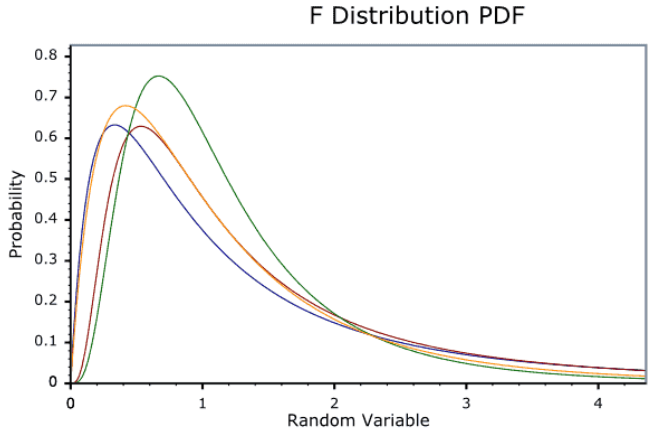
### Mean absolute error (MAE)

Average absolute prediction error.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} | y_i - \hat{y}_i |$$

All three are expressed in the units of $y$.
Smaller values indicate better fit.

# Assessing the model: model usefulness




## Overall F-test

Test whether the regression explains anything beyond the mean-only model:

$$H_0: \beta_1 = 0$$

F statistic:

$$F = \frac{(\mathrm{TSS} - \mathrm{RSS})/(p-1)}{\mathrm{RSS}/(n-p)} = \frac{\mathrm{ESS}/(p-1)}{\mathrm{RSS}/(n-p)}$$

Under $H_0$: $F \sim F_{(p-1)(n-p)}$

- $\mathrm{ESS}/(p-1)$ is the **average explained variability per parameter** you added beyond the intercept.
- $\mathrm{RSS}/(n-p)$ is the **average unexplained variability per residual degree of freedom** (this is essentially $\hat{\sigma}^2$)

So: **F compares "signal per parameter" to "noise per degree of freedom."**

- If the numerator is large relative to the denominator → the model is explaining much more than what noise would produce → **evidence against the null**.
- If the ratio is close to 1 → the improvement looks like noise-level fluctuation → **no strong evidence**.

# Inference about Y

## Confidence interval for the mean response

Target:

$$\mu_{Y|X=x_0} := \mathbb{E}[Y \mid X = x_0]$$

Question it answers: *"What is the average exam score of students who study $x_0$ hours?"*

We estimate $\mu_{Y|X=x_0}$ by $\hat{y}_0$, and we need a CI because $\hat{\beta}_0, \hat{\beta}_1$ are estimated from finite data → <u>the fitted line is uncertain</u>.

$$\mathrm{Var}(\hat{y}_0) = \mathrm{Var}\left(\hat{\beta}_0 + \hat{\beta}_1 x_0\right)$$

Using the variance rule for a linear combination:

$$\mathrm{Var}(A + B) = \mathrm{Var}(A) + \mathrm{Var}(B) + 2\mathrm{Cov}(A, B)$$

we get

$$\mathrm{Var}(\hat{y}_0) = \mathrm{Var}\left(\hat{\beta}_0\right) + x_0^2 \mathrm{Var}\left(\hat{\beta}_1\right) + 2x_0 \mathrm{Cov}\left(\hat{\beta}_0, \hat{\beta}_1\right)$$

If you plug in the standard simple linear regression formulas for these quantities and simplify, you obtain

$$\mathrm{Var}(\hat{y}_0) = \sigma^2 \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

So the standard error is

$$\mathrm{SE}(\hat{y}_0) = \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$$

# Inference about Y

## Confidence interval for the mean response

Target:

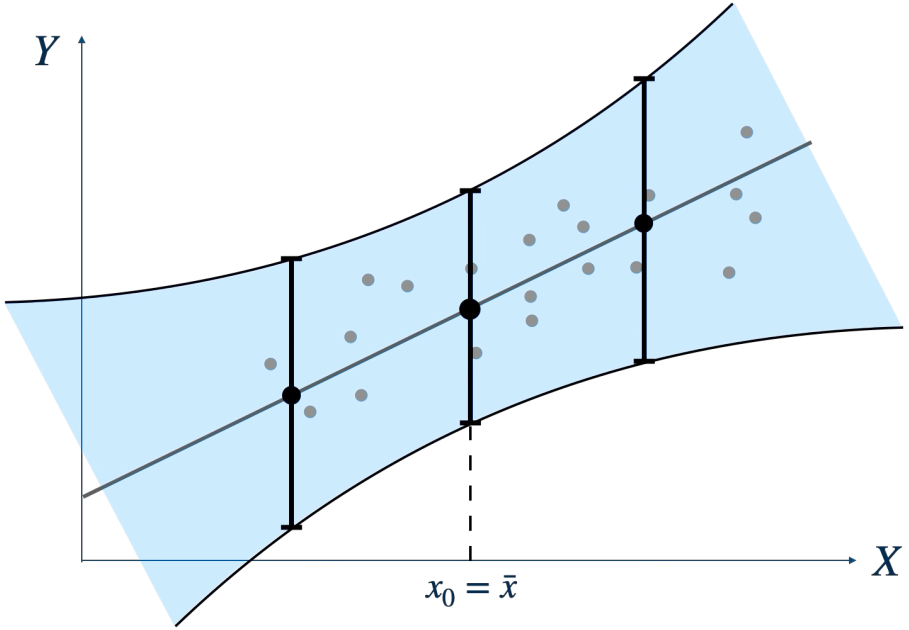
$$\mu_{Y|X=x_0} := \mathbb{E}[Y \mid X = x_0]$$

Question it answers: *"What is the average exam score of students who study $x_0$ hours?"*

We estimate $\mu_{Y|X=x_0}$ by $\hat{y}_0$, and we need a CI because $\hat{\beta}_0, \hat{\beta}_1$ are estimated from finite data → <u>the fitted line is uncertain</u>.

A $100(1-\alpha)\%$ CI is:

$$\hat{y}_0 \pm t_{n-2,1-\alpha/2}\, \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$$




- the interval is narrowest near $x_0 = \bar{x}$
- it widens as $x_0$ moves away from $\bar{x}$ (extrapolation is uncertain)
- it reflects **uncertainty about the mean line** (parameter uncertainty)

# Inference about Y

## Prediction interval for a new observation

Target:

$$Y_{new} \mid X = x_0$$

Question it answers: "If one new student studies $x_0$ hours, what range of scores could we observe?"

This must include:

- uncertainty about the mean line (same as above)
- **irreducible individual variability** around the mean (the noise $\varepsilon$)

So the prediction interval is:

$$\hat{y}_{\text{new}} \pm t_{n-2,1-\alpha/2}\, \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^{n}(x_0 - \bar{x})^2}}$$

That "+1" is the variance of the new error term $\varepsilon_{new}$.
So:

- **CI for mean**: uncertainty of the estimated mean response
- **PI for new Y**: uncertainty of mean + unavoidable noise → always wider

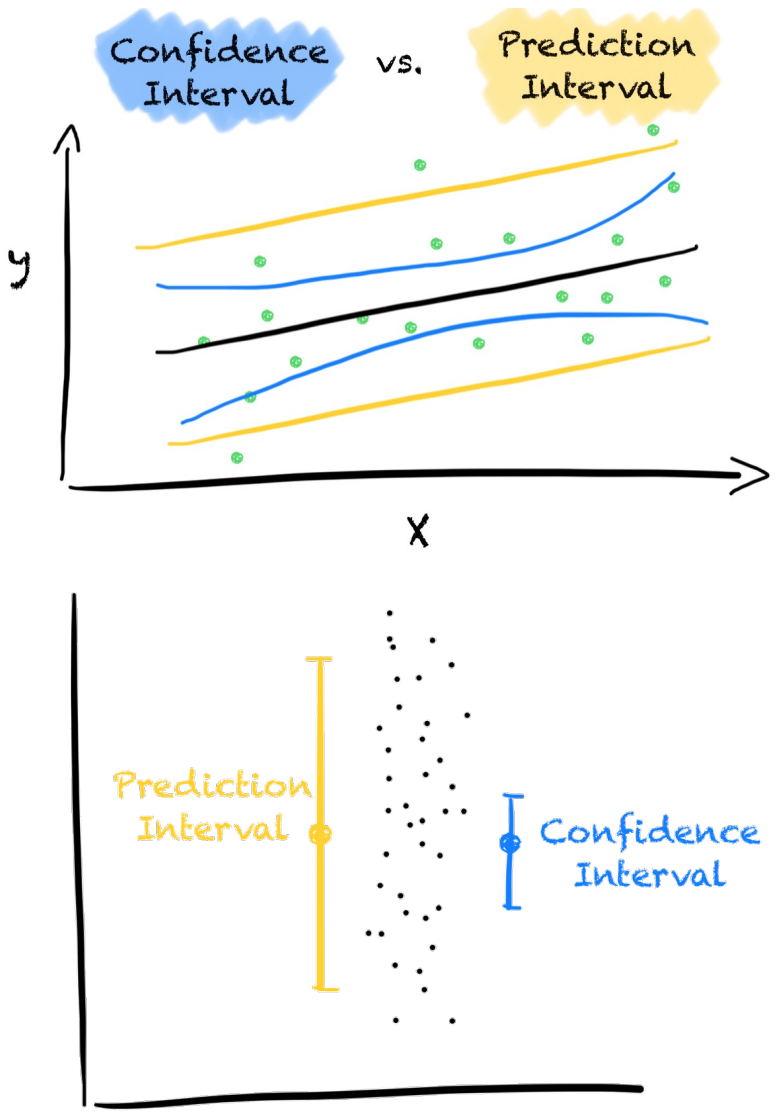
# Confidence vs Prediction Intervals

**Confidence Interval**

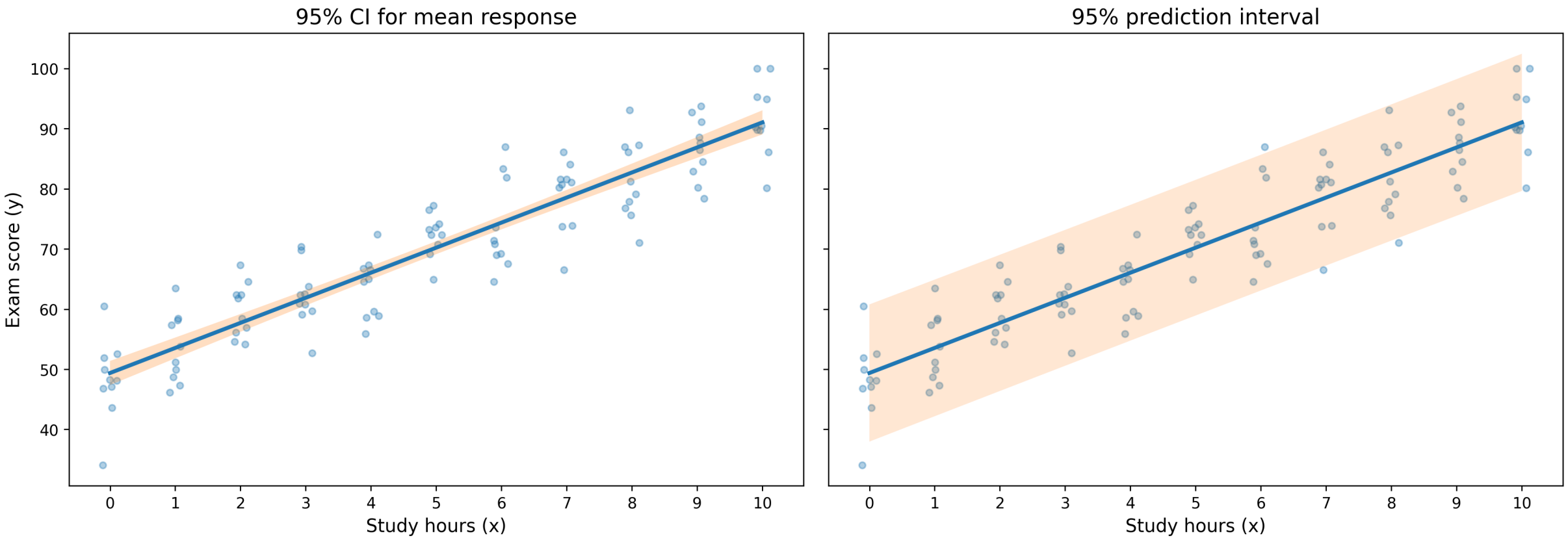
- shows the **uncertainty of a population parameter**, such as the mean or a regression coefficient.
- The sample standard deviation being larger, results in wider Confidence Intervals. Due to the higher variation, we can be less certain that the sampled parameter is close to the population parameter.
- A larger sample size can balance the effect of a few outliers while the samples are more similar. Hence, we can be more certain and thus have a narrower Confidence Interval.

**Prediction Interval**

- shows the **uncertainty of a specific value**. ("We are 95% confident that the next observation will fall within this range.")
- focuses on prediction of new events



# Confidence vs Prediction Intervals



# Appendix

## Useful material

**Classic Textboox (free download):**

- Kutner et al. (2005) – Applied Linear Statistical Models, Chapter 1. https://users.stat.ufl.edu/~winner/sta4211/ALSM_5Ed_Kutner.pdf
- James, G., Witten, D., Hastie, T., Tibshirani, R. (2013) - *An Introduction to Statistical Learning*. Springer, Chapter on Linear Regression https://www.statlearning.com/

**Course notes:**

- Penn State University STAT 501, Lesson 1 – https://online.stat.psu.edu/stat501/lesson/1
- OpenIntro Statistics, Chapter 8 - https://leanpub.com/os